nature medicine

Article

https://doi.org/10.1038/s41591-023-02677-x

Toripalimab plus nab-paclitaxel in metastatic or recurrent triple-negative breast cancer: a randomized phase 3 trial

In the format provided by the authors and unedited

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Supplemental Table 1. Summary of reasons for screening failure

	Total screened patients N=799
Screening failure	268 (33.5)
Reasons for screening failure	
Not meeting Inclusion or meeting exclusion criteria	232 (86.6)
Withdraw of consent	33 (12.3)
Did not complete all tests within the screening period	3 (1.1)

Supplemental Table 2. Common reasons (≥5 incidences) for not meeting inclusion criteria or meeting exclusion criteria.

Code	Inclusion or exclusion criteria details	N
EXCL03A	Patients with active or untreated brain metastasis determined by magnetic resonance imaging (MRI) evaluation during the screening period. Patients with CNS metastases stable for ≥1 month after local treatment, and who had stopped systemic hormone therapy (> 10 mg /d prednisone or equivalent) for >4 weeks before randomization can participate in the study.	49
INCL11A	Laboratory values within 1 week prior to the first dose of the investigational drugs must be within the specified range.	40
INCL06A	Patients initially diagnosed with stage IV (based on AJCC, 8th edition) or with recurrent/metastatic TNBC who are not amenable for surgical treatment. Prior antitumor therapies such as taxanes during neoadjuvant and/or adjuvant therapy are allowed, provided that no disease progress occurred after neoadjuvant therapy, and that the interval between the end of (neo)adjuvant therapy of taxanes and recurrence/metastasis was ≥12 months or the interval between the end of (neo)adjuvant therapy of capecitabine, etc., and recurrence/metastasis was ≥6 months.	21
INCL03A	Histologically confirmed TNBC, namely HER2, ER and PR were negative.	20
EXCL21A	Patients with active hepatitis B. Patients with previous HBV infection or cured hepatitis B may be enrolled when satisfying the following conditions: positive HBcAb and negative HBsAg, the number of HBV DNA copies less than the upper limit of normal of laboratory department of study site before randomization.	19
EXCL30	Any other severe medical diseases or clinical laboratory test abnormalities that may prevent the patients from safe participation and completion of the study as judged by the investigator.	13
EXCL20A	Patients with any active autoimmune disease, requiring systemic therapy (i.e. use of disease-regulating drugs, corticosteroids or immunosuppressive drugs) in the past 2 years. Hormone replacement therapy (such as thyroxine, insulin, or physiologic corticosteroid replacement therapy for adrenal or pituitary insufficiency) is not considered as systemic therapy.	12
INCL04A	Presence of at least one measurable lesion according to the definitions in RECIST v1.1 criteria. For the lesion of previous radiotherapy, only when the lesion has clear disease progression after radiotherapy, it may be identified as a measurable lesion.	10
EXCL10A	Patients with malignant tumors within ≤ 5 years prior to the first dose of the investigational drugs. Exceptions for patients with radical early malignant tumor (carcinoma in situ or stage I tumor), e.g. adequately treated cervical carcinoma in situ, basal or squamous cell skin cancer.	6
INCL07A	Patients who are able to provide tumor tissue blocks and/or unstained pathological sections, due to newly acquired biopsy tissues being preferred over archived tissues.	6
EXCL05A	Uncontrolled pleural effusion, peritoneal effusion, or pericardial effusion (patients with indwelling drainage tube are allowed to be enrolled to this study).	5
EXCL14A	Suffering severe cardiovascular disease, such as myocardial infarction, unstable arrhythmia or unstable angina within 3 months prior to randomization, with an	5

	electrocardiogram (ECG) exception that is considered clinically significant by the investigator.	

Supplemental Table 3. PD-L1 positive TNBC prevalence determined by 22C3 and JS311 IHC assays.

PD-L1 IHC staining was performed on 103 TNBC biopsy samples by JS311 (Ventana BenchMark ULTRA) and 22C3 (Darko Autostainer Link 48) IHC assays. The combined positive score (CPS) of JS311 assay was calculated by dividing the number of PD-L1-positive cells, including tumor cells, lymphocytes, macrophages, dendritic cells and granulocytes to the total number of all tumor cells and immune cells in all tumor-associated areas and multiplying by 100.

Cutoff	JS311	22C3
CPS ≥ 1	58 (56.3%)	46 (44.7%)
CPS ≥ 5	41 (39.8%)	28 (27.2%)
CPS ≥ 10	27 (26.2%)	14 (13.6%)

Supplemental Table 4. TNM classification for the ITT population.

	Toripalimab + chemo (N=353) N (%)	Placebo + chemo (N=178) N (%)	Total (N=531) N (%)
TNM Grade primary tumor（T）			
Tx	113 (32.0)	54 (30.3)	167 (31.5)
T0	59 (16.7)	31 (17.4)	90 (16.9)
Tis	0	0	0
T1/T1mi/T1a/T1b/T1c	26 (7.4)	11 (6.2)	37 (7.0)
T2	88 (24.9)	42 (23.6)	130 (24.5)
T3	17 (4.8)	12 (6.7)	29 (5.5)
T4/T4a/T4b/T4c/T4d	50 (14.2)	28 (15.7)	78 (14.7)
TNM Grade regional LN（N）			
Nx	93 (26.3)	39 (21.9)	132 (24.9)
N0	78 (22.1)	35 (19.7)	113 (21.3)
N1/N1mi	50 (14.2)	22 (12.4)	72 (13.6)
N2/N2a/N2b	38 (10.8)	28 (15.7)	66 (12.4)
N3/N3a/N3b/N3c	94 (26.6)	54 (30.3)	148 (27.9)
TNM Grade Metastasis (M)			
M0	18 (5.1)	15 (8.4)	33 (6.2)
M1	335 (94.9)	163 (91.6)	498 (93.8)

Supplemental Table 5. TNM classification for the PD-L1 positive subgroup.

	Toripalimab + chemo (N=200) N (%)	Placebo + chemo (N=100) N (%)	Total (N=300) N (%)
TNM Grade primary tumor（T）			
Tx	54 (27.0)	28 (28.0)	82 (27.3)
T0	38 (19.0)	18 (18.0)	56 (18.7)
Tis	0	0	0
T1/T1mi/T1a/T1b/T1c	17 (8.5)	6 (6.0)	23 (7.7)
T2	51 (25.5)	21 (21.0)	72 (24.0)
T3	10 (5.0)	9 (9.0)	19 (6.3)
T4/T4a/T4b/T4c/T4d	30 (15.0)	18 (18.0)	48 (16.0)
TNM Grade regional LN（N）			
Nx	46 (23.0)	20 (20.0)	66 (22.0)
N0	38 (19.0)	22 (22.0)	60 (20.0)
N1/N1mi	23 (11.5)	12 (12.0)	35 (11.7)
N2/N2a/N2b	27 (13.5)	16 (16.0)	43 (14.3)
N3/N3a/N3b/N3c	66 (33.0)	30 (30.0)	96 (32.0)
TNM Grade Metastasis (M)			
M0	10 (5.0)	9 (9.0)	19 (6.3)
M1	190 (95.0)	91 (91.0)	281 (93.7)

Supplemental Table 6. Subsequent Anti-Cancer Therapy in Intent-to-Treat Population After the Study Treatment.

	Toripalimab + Chemo N = 353 (%)	Placebo + Chemo N= 178 (%)	Total N=531 (%)
New anti-cancer therapy			
Systemic treatment	186 (52.7)	94 (52.8)	280 (52.7)
Surgical treatment	14 (4.0)	11 (6.2)	25 (4.7)
Radiotherapy	42 (11.9)	31 (17.4)	73 (13.7)
Other	35 (9.9)	17 (9.6)	52 (9.8)
Systemic treatment			
PD-1/L1 Inhibitors	21 (5.9)	11 (6.2)	32 (6.0)
Cytotoxic Therapy	174 (49.3)	89 (50.0)	263 (49.5)
Targeting Therapy	75 (21.2)	31 (17.4)	106 (20.0)
Other	5 (1.4)	3 (1.7)	8 (1.5)

Data Cutoff: November 30, 2022

Supplemental Table 7. Objective response rate (INV-ORR), disease control rate (INV-DCR) and duration of response (INV-DoR) assessed by the investigator in accordance with RECIST v1.1 in the PD-L1+ Population at the interim PFS analysis.

Variable	Toripalimab + chemo (N=200)	Placebo +chemo (N=100)
Best overall response - no. (%)		
Complete response	9 (4.5)	3 (3.0)
Partial response	123 (61.5)	63 (63.0)
Stable disease	44 (22.0)	22 (22.0)
Progressive disease	15 (7.5)	8 (8.0)
Not evaluable or missing	9 (4.5)	4 (4.0)
Objective response rate (ORR)		
ORR % (95% CI)	66 (59.0, 72.5)	66 (55.8, 75.2)
Difference in ORR % (95% CI)	-0.3 (-11.7, 11.0)	
P-value[1]	0.9549	
Disease control rate (DCR)		
DCR % (95% CI)	88 (82.7, 92.2)	88 (80.0, 93.6)
Difference in DCR % (95% CI)	-0.6 (-8.3, 7.1)	
P-value[1]	0.8796	
Estimate of median DoR (95% CI) (months)	11.1(7.2, 13.8)	4.4(4.2, 5.6)
Stratified estimate of hazard ratio (toripalimab versus placebo) (95%CI) [2]	0.507(0.346, 0.744)	
P-value[3]	0.0004	

Data Cutoff: November 30, 2022

CR, complete response; PD, progressive disease; NE, not estimable.

[1]In the stratified analysis, nominal p-value (two-sided) of the comparison between arms was calculated using Cochran-Mantel-Haenszel method based on random stratification factors, and the 95% confidence interval for the difference in the rate difference between the arms was estimated based on Mantel-Haenszel, stratified by PD-L1 expression status, prior paclitaxel therapy, and current line of therapy.

[2]HR for DoR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were PD-L1 expression status, prior paclitaxel therapy, and current line of therapy.

[3]P-value (two-sided) for DoR was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 8. Objective response rate (INV-ORR), disease control rate (INV-DCR) and duration of response (INV-DoR) assessed by the investigator in accordance with RECIST v1.1 in the Intent-to-Treat Population at the interim PFS analysis.

Variable	Toripalimab + chemo (N=353)	Placebo +chemo (N=178)
Best overall response - no. (%)		
Complete response	14 (4.0)	3 (1.7)
Partial response	201 (56.9)	99 (55.6)
Stable disease	87 (24.6)	53 (29.8)
Progressive disease	33 (9.3)	14 (7.9)
Not evaluable or missing	18 (5.1)	9 (5.1)
Objective response rate (ORR)		
ORR % (95% CI)	60.9 (55.6, 66.0)	57.3 (49.7, 64.7)
Difference in ORR % (95% CI)	3.3 (-5.5, 12.0)	
P-value[1]	0.4682	
Disease control rate (DCR)		
DCR % (95% CI)	85.6 (81.4, 89.0)	87.1(81.2, 91.6)
Difference in DCR % (95% CI)	-1.6 (-7.7, 4.6)	
P-value[1]	0.6219	
Estimate of median DoR (95% CI) (months)	8.5(6.9, 11.3)	5.6(4.2, 7.4)
Stratified estimate of hazard ratio (toripalimab versus placebo) (95%CI) [2]	0.591(0.437, 0.801)	
P-value[3]	0.0006	

Data Cutoff: November 30, 2022

CR, complete response; PD, progressive disease; NE, not estimable.

[1]In the stratified analysis, nominal p-value (two-sided) of the comparison between arms was calculated using Cochran-Mantel-Haenszel method based on random stratification factors, and the 95% confidence interval for the difference in the rate difference between the arms was estimated based on Mantel-Haenszel, stratified by PD-L1 expression status, prior paclitaxel therapy, and current line of therapy.

[2]HR for DoR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were PD-L1 expression status, prior paclitaxel therapy, and current line of therapy.

[3]P-value (two-sided) for DoR was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 9. Study Drug Exposure.

	ITT		PD-L1+	
	Toripalimab + Chemotherapy N=353	**Placebo + Chemotherapy N=175**	**Toripalimab + Chemotherapy N=200**	**Placebo + Chemotherapy N=100**
Toripalimab/ Placebo				
Actual treatment duration, mean, weeks	29.194	28.308	31.040	27.749
median(range), weeks	21.143(0.143-129.429)	22.143(0.143-125.571)	22.071(0.143-129.429)	22.286(0.143-125.571)
Actual cycle received, mean	9.6	9.9	10.0	9.5
median(range)	8.0 (1-39)	8.0 (1-36)	8.0 (1-39)	8.0 (1-35)
Nab-Paclitaxel				
Actual treatment duration, mean, weeks	27.634	27.892	29.984	26.430
median(range), weeks	22.143(0.143-117.0)	22.429(0.143-125.571)	22.929(0.143-117.0)	22.429(0.143-125.571)
Actual cycle received, mean	9.0	9.4	9.6	8.8
median(range)	8.0 (1-36)	8.0 (1-34)	8.0 (1-36)	8.0 (1-34)

NA=not applicable; Nab = Nanoparticle albumin bound. Three PD-L1 negative/non-evaluable patients in the placebo arm did not receive any study treatment and were excluded from safety dataset.

Data Cutoff: November 30, 2022

Supplemental Table 10. Treatment-Emergent Adverse Events Reported in At Least 10% of the Patients in either treatment arm

	Toripalimab + nab-paclitaxel N = 353		Placebo + nab-paclitaxel N= 178	
System Organ Class Preferred Term	**All Grades**	**Grades ≥3**	**All Grades**	**Grade ≥3**
Any TEAE (%)	99.2	56.4	98.9	54.3
Blood and lymphatic system disorders				
Leukopenia	72.8	25.2	75.4	23.4
Neutropenia	67.7	26.3	64.0	28.6
Anemia	60.9	4.0	56.6	2.9
Lymphopenia	13.0	3.1	14.9	3.4
Thrombocytopenia	9.9	0.6	10.9	0.0
Investigations				
Aspartate aminotransferase increased	45.6	6.5	49.1	1.7
Alanine aminotransferase increased	44.8	5.9	48.0	2.3
Gamma-glutamyl transferase increased	15.6	2.8	12.0	0.6
Blood bilirubin increased	13.3	0.0	13.1	1.1
Blood alkaline phosphatase increased	13.0	0.3	10.9	0.6
Weight decreased	11.3	0.8	9.1	0.0
Blood lactate dehydrogenase increased	10.8	0.3	9.1	0.0
Skin and subcutaneous tissue disorders				
Alopecia	67.4	0	61.7	0
Rash	22.1	2.0	22.9	0.0
Metabolism and nutrition disorders				
Hypertriglyceridemia	25.8	1.7	29.7	1.1
Decreased appetite	21.0	0.8	15.4	1.1
Hyperuricemia	14.7	0.0	15.4	0.6
Hypoalbuminemia	14.7	0.0	14.3	0.0
Hypokalemia	13.9	2.3	9.1	2.3
Hyperglycemia	12.7	0.3	16.0	0.0
Hypercholesterolemia	12.5	0.3	16.6	0.0
Hyponatremia	10.2	0.6	10.9	0.6
General disorders and administration site conditions				
Asthenia	31.4	4.2	21.7	2.3
Pyrexia	17.0	0.0	13.1	0.0
Peripheral edema	14.4	0.3	22.3	0.6

System Organ Class Preferred Term	Toripalimab + nab-paclitaxel N = 353		Placebo + nab-paclitaxel N= 178	
	All Grades	**Grades ≥3**	**All Grades**	**Grade ≥3**
Gastrointestinal disorders				
Nausea	26.6	1.1	15.4	0.6
Constipation	21.2	0.0	16.0	0.6
Vomiting	19.5	0.8	12.0	0.0
Diarrhea	14.2	1.1	12.6	0.6
Nervous system disorders				
Hypoesthesia	21.2	4.5	13.7	1.7
Neurotoxicity	13.0	4.2	14.3	4.0
Insomnia	8.8	0.3	10.3	0
Infections and infestations				
Urinary tract infection	15.9	1.1	16.0	0.6
Endocrine disorders				
Hypothyroidism	19.0	0.0	8.0	0.0
Respiratory, thoracic and mediastinal disorders				
Cough	10.2	0.0	11.4	0.0

TEAE = treatment-emergent adverse event. Note: Adverse event coding version: MedDRA 25.1.

Note: Adverse events that occurred during treatment include all adverse events that occurred between the first dose and 30 days after the last dose or before starting a new anti-cancer therapy, whichever occurred first.

If a patient experiences the same adverse event multiple times, that patient is counted only once under the same system organ classification and preferred terminology when calculating the number of adverse events.

Data Cutoff: November 30, 2022

Supplemental Table 11. Immune related adverse events (irAE) reported in more than one patient in either arm as assessed by the investigator.

No. of patients (%)	Toripalimab + chemo N=353 (%)		Placebo + chemo N=175 (%)	
	All Grades	Grades ≥3	All Grades	Grade ≥3
Any irAE				
Hypothyroidism	13.3	0.0	4.0	0.0
Rash	9.3	1.7	8.6	0.0
Hyperthyroidism	7.1	0.0	1.1	0.0
Aspartate aminotransferase increased	5.1	2.8	3.4	0.6
Alanine aminotransferase increased	4.5	2.5	2.9	0.6
Asthenia	3.4	1.1	2.3	0.6
Pneumonitis	2.8	0.6	0.0	0.0
Nausea	2.5	0.6	0.6	0.0
Neutrophil count decreased	2.3	1.1	1.1	1.1
Pruritus	2.3	0.0	1.7	0.0
Decreased appetite	2.3	0.0	1.7	0.0
Amylase increased	2.0	0.3	0.6	0.6
White blood cell count decreased	2.0	1.1	1.1	1.1
Gamma-glutamyl transferase increased	1.7	0.8	0.0	0.0
Immune-mediated lung disease	1.7	0.8	0.6	0.6
Oedema peripheral	1.7	0.3	2.9	0.6
Hypoesthesia	1.7	0.0	0.6	0.0
Blood bilirubin increased	1.4	0.0	0.6	0.0
Lipase increased	1.4	1.1	0.0	0.0
Hypertriglyceridemia	1.4	0.0	0.0	0.0
Cough	1.4	0.0	1.1	0.0
Pyrexia	1.4	0.0	1.7	0.0
Constipation	1.4	0.0	0.0	0.0
Diabetes mellitus	1.1	0.8	0.0	0.0
Pain in extremity	1.1	0.0	0.6	0.0
Immune-mediated hepatitis	1.1	0.8	0.0	0.0
Anemia	1.1	0.0	1.7	0.0
Immune-mediated thyroiditis	0.8	0.0	0.0	0.0
Thyroiditis	0.8	0.0	0.0	0.0
Bilirubin conjugated increased	0.8	0.3	0.6	0.0
Blood creatinine increased	0.8	0.0	0.6	0.0
Blood thyroid stimulating hormone increased	0.8	0.0	1.7	0.0
Erythema multiforme	0.8	0.3	0.0	0.0
Hyperlipidemia	0.8	0.0	0.0	0.0
Face oedema	0.8	0.0	0.0	0.0
Peripheral sensory neuropathy	0.8	0.0	1.1	0.0
Rash pustular	0.8	0.0	0.6	0.0

No. of patients (%)	Toripalimab + chemo N=353 (%)		Placebo + chemo N=175 (%)	
	All Grades	Grades ≥3	All Grades	Grade ≥3
Autoimmune thyroiditis	0.6	0.0	0.0	0.0
Hypopituitarism	0.6	0.3	0.0	0.0
Blood alkaline phosphatase increased	0.6	0.0	0.0	0.0
Platelet count decreased	0.6	0.3	1.1	0.0
Dermatitis	0.6	0.0	0.6	0.6
Immune-mediated dermatitis	0.6	0.0	0.0	0.0
Hyperglycemia	0.6	0.0	2.9	0.0
Hypoalbuminemia	0.6	0.0	1.1	0.0
Hypophosphatemia	0.6	0.0	0.6	0.0
Interstitial lung disease	0.6	0.3	0.6	0.0
Influenza like illness	0.6	0.0	1.1	0.0
Vomiting	0.6	0.3	0.6	0.0
Dizziness	0.6	0.0	0.0	0.0
Neurotoxicity	0.6	0.6	0.0	0.0
Conjunctivitis	0.6	0.0	0.0	0.0
Arthralgia	0.6	0.0	0.6	0.0
Eyelid oedema	0.6	0.0	0.6	0.0
Renal impairment	0.6	0.3	0.0	0.0
Abdominal distension	0.3	0.0	1.1	0.0
Diarrhea	0.3	0.0	1.1	0.0
Proteinuria	0.3	0.0	1.1	0.0

Note: For the same patient if the same adverse event occurs multiple times, that patient is counted only once in the same system organ classification and preferred terminology when calculating the number of adverse events. Three PD-L1 negative/non-evaluable patients in the placebo arm did not receive any study treatment and were excluded from safety dataset.

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